



Programming Fundamentals
with JavaScript



What is a programming language?

It is a **formal language** that specifies a group of **input instructions** that can be used to produce **an output**



formal

language

a series of symbols that respect grammatical and syntactical rules



input

instructions

by knowing the rules we can generate instructions to be executed



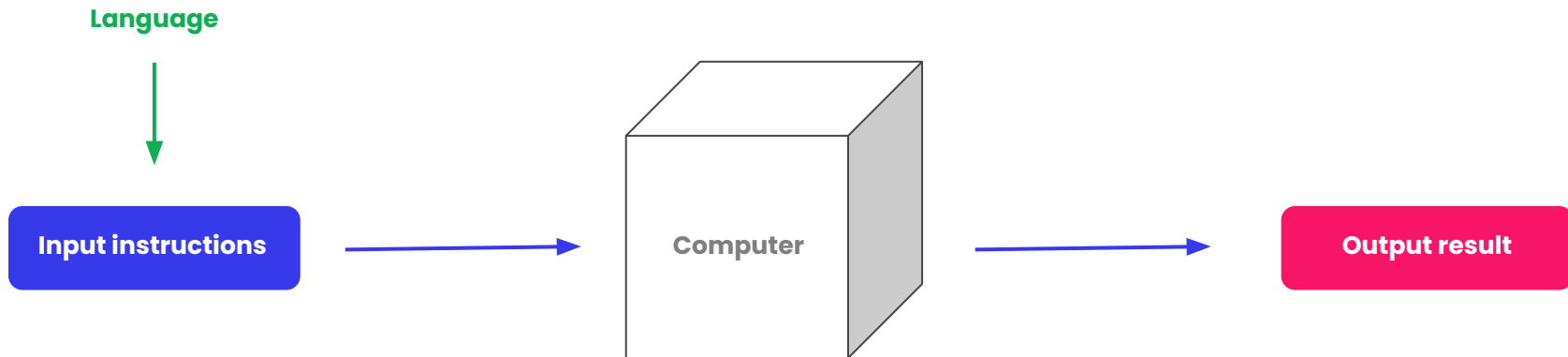
an

output

by following our instructions, the computer can produce a visible output



What is a programming language?

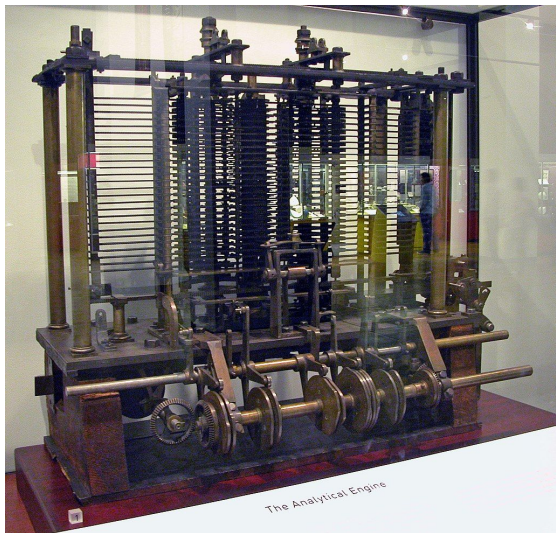




The analytical machine

A bit of history – how we got to where we are today

What now appears to be a simple concept was *definitely* not in the 19th century:
in full industrial revolution, specialized machines were built following the dictates of Taylorism (and then Fordism)



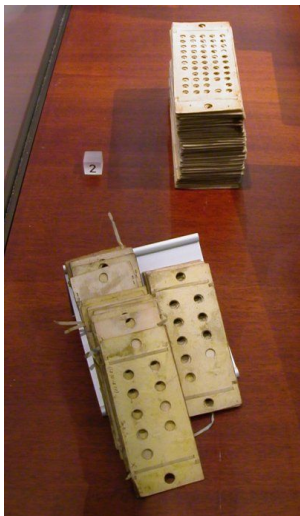
In **1837**, **Charles Babbage** designed **a very complicated machine** the size of a room and powered by a large steam chamber, a machine that was not specialized but could be **programmed** by a technician to perform always different calculations, with the following characteristics:

- An **instructional language** created by **Ada Lovelace** (the first programmer in the world)
- A system for **giving instructions** through **punch cards**
- An **output system** consisting of a **printer** and a **series of needles** to make graphs to write the result



Punch cards

How code was once written



Punch cards for the analytical machine



Programmers using a punch card machine in 1970



Programming languages

Programming languages are a **convenient** way for us **humans** to deal with those 1s and 0s and tell the computer what we'd like it to do

According to Wikipedia there's currently **over 700** programming languages available

We can group them into 2 main categories (kinda):

- **Compiled** (turning code to 1s and 0s before the program can run)
- **Interpreted** (turning code to 1s and 0s as the program is running)

Compiled

- ADA
- C
- C++
- Go
- Rust

Interpreted

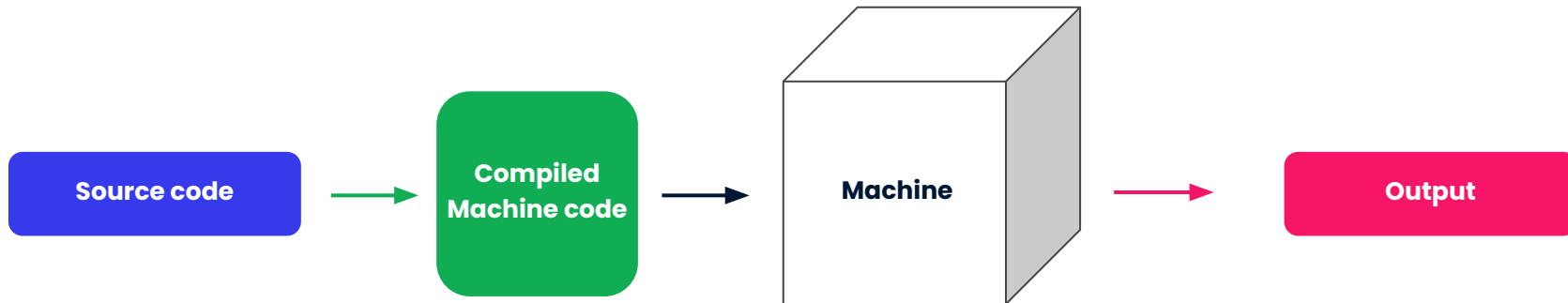
- PHP
- Javascript
- Ruby
- Basic
- Python



Programming languages

Compiled languages

The instructions written by the programmer, or **source code**, is given to a program called a **compiler**, which translates the code into **machine code** (binary!) ahead of time and produces an **executable** file that the current machine can run

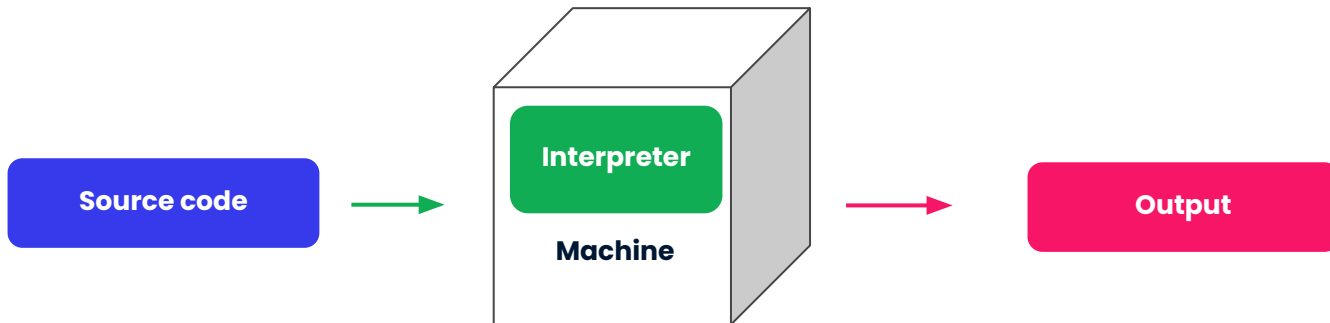




Programming languages

Interpreted languages

The **source code** here is given to a program called an **interpreter** that runs it line by line and tells the machine what to do **in real time**, with **no compilation** step required, but generally **much slower** than compiled code





Programming languages

So, which is better?

Well, allow me to introduce the answer that you will very likely hear (and say) a million times (sometimes per day) as a developer:

It depends

Like many things in the programming world, there is no right or wrong, just **trade-offs**

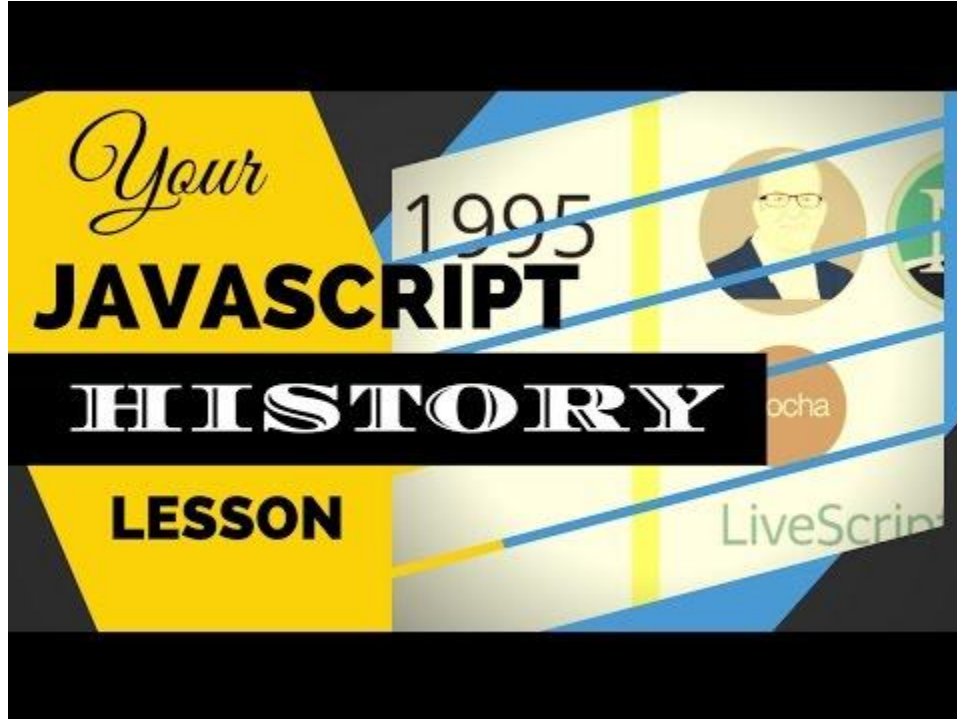
However, we can say that, in general, **compiled languages perform better** than their interpreted counterparts

On the other hand, **interpreted languages are more portable**, i.e. they work on multiple operating systems without having to be rewritten or recompiled, and can be modified while they are running

Also, as you will learn, some languages live somewhere between interpretation and compilation



JavaScript





JavaScript

Today JavaScript is **everywhere**! You use it to program things from **toasters** to **spaceship apps**, **frontends**, **backends** and pretty much anything you can think of

However, it is still the only language that is natively run by browsers, and so, it allows you to do some unique things, like:

- Create, read, update and delete **elements** on the page
- React to user **events** (clicks, scrolls, etc)
- Validate **data** entered by the user
- Check which browser is being used
- Check the current size of the screen
- And more...

These features allow us to create **entire applications** within the browser.

But for now, we're going to focus on **Node.js**



Browser v Node.js

